

# GABRIELLE WIDJAJA

## DATA SCIENTIST

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## SUMMARY

Data scientist with experience in machine learning, deep learning, and NLP, and a strong background in bioinformatics and data analytics. Proficient in Python, SQL, and R, with hands-on work applying AI/ML to real-world problems. Currently pursuing an M.S. in Computational Data Analytics at Georgia Tech, with a focus on machine learning and unstructured data. Skilled in building predictive models, analyzing large datasets, and deploying data-driven solutions.

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## EXPERIENCE

### **Children's Hospital of Philadelphia, Philadelphia, PA**

*Sequencing Core Research Technician II (October 2022 – Current)*

- Built a robust data analysis pipeline in R and Python to clean, normalize, and analyze gene expression data from over 20,000 genes, enabling hypothesis testing, statistical validation, and downstream pathway analysis.
- Developed interactive dashboards and visualizations in Python and R to communicate analytical insights and results clearly to research stakeholders.

### **NASA-Affiliated Multi-Omics/Systems Biology Analysis Working Group (AWG)**

*Contributor, Data Scientist (January 2023 – Current)*

- Developed a custom gene list through data cleanup and filtering of large multi-omics datasets to assess bioenergetic and mitochondrial-immune alterations; applied GSEA models to extract biological insights.
- Created complex visualizations to illustrate key findings, several of which were featured in consortium publications; actively contributed to COVID-19 and spaceflight-related studies as part of the COV-IRT and NASA-affiliated Multi-Omics/Systems Biology Analysis Working Group (AWG).

### **Temple University Ambler Field Station**

*Research Intern (Dr. Freestone) (August 2021 – May 2022)*

- Developed predictive models to assess the impact of wind disturbances on biodiversity.
  - Conducted statistical analysis on ecological datasets using Python and R.
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## PROJECTS

### **Telecommunications Customer Engagement to Adoption – Course Project:**

- Collaborated with a student team to analyze customer engagement data from Calix, using Python and Snowflake in an Azure virtual desktop (AVD) environment to clean, normalize, and engineer features for predictive modeling and causal analysis on product adoption.
- Presented midterm findings and actionable insights to Calix's Lead Business Analyst, showcasing data correlations between feature usage and customer adoption, with final report and recommendations in progress.

### **Optimizing Insurance Premiums & Reducing Financial Risks – Course Project:**

- Applied SVM, KNN, logistic regression, random forest, and gaussian mixture models to segment populations by demographic and clinical factors for risk-based insurance premium tiering.
- Visualized and analyzed model performance to support team decision-making on model selection; collaborated to evaluate trade-offs and presented final recommendations and implementation considerations as part of a course capstone project.

### **ML- Driven Interactive Crime Mapping for NYC – Course Project:**

- Used DBSCAN and Random Forest to analyze spatial, temporal, demographic, and crime trends in NYC, and to predict areas with elevated crime rates.

- Used Tableau for crime forecasting and dynamic data visualization, enabling data-driven decision-making for law enforcement and businesses.

#### **Pandemic Flu Spread Simulation – Course Project:**

- Developed a modified SVEIRD model in R to simulate virus transmission dynamics and evaluate the impact of interventions such as masking, vaccination, and quarantine.
- Conducted extensive simulations to optimize public health strategies, revealing that quarantining reduced lethality, vaccination lowered deaths per infection, and masking had minimal impact on lethality, providing data-driven insights for policymakers.

#### **Patent Analysis and Machine Learning Experiment – Course Project:**

- Applied NLP and ML techniques to analyze patent data, leveraging scikit-learn, NLTK, and LDA/NMF topic modeling to extract key insights from patent abstracts.
- Developed a document-term matrix (DTM), performed Lasso regression to evaluate topic informativeness, and built predictive models using Decision Tree, Random Forest, SGD, and LightGBM, optimizing hyperparameters with five-fold cross-validation and macro F1-score tuning.

#### **Space Travel & Mitochondrial Research:**

- Contributed to multi-omics data integration for NASA research investigating the effects of long-term space travel.

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## SKILLS

**Machine Learning & AI** – Classification, regression, clustering, ranking, random forest, XGBoost, TensorFlow, PyTorch, transformers, LLMs.

**Data Processing & NLP** – Text processing, sentiment analysis, NER, topic modeling (LDA, NMF), Word2Vec, BERT, GPT.

**Programming & Cloud Computing** – Python, SQL, R, AWS, GCP, Azure, Spark, model deployment.

**Data Visualization & Statistical Analysis** – Tableau, Power BI, Matplotlib, Seaborn, hypothesis testing, anomaly detection.

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## EDUCATION

**Georgia Institute of Technology, Atlanta GA**

*M.S. Computational Data Science (Expected Graduation Date: December 2025)*

#### **Coursework:**

Intro to Analytics Modeling, Business Fundamentals for Analytics, Computing for Data Analysis, Data Analytics in Business, Regression Analysis, Simulation, Analysis of Unstructured Data, Data and Visual Analytics

**Temple University, Philadelphia PA**

*B.S. Biochemistry*

#### **Achievements:**

Temple Emerging STEM Scholars, Undergraduate Research Program

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## PUBLICATIONS

- Guarnieri JW, Lie T, Albrecht YES, Hewins P, Jurado KA, Widjaja GA, Zhu Y, McManus MJ, Kilbaugh TJ, Keith K, Potluri P, Taylor D, Angelin A, Murdock DG, Wallace DC. Mitochondrial Antioxidants Abate SARS-COV-2 Pathology in Mice. *Proc Natl Acad Sci.* 2024; 121 (30) e2321972121
- Guarnieri JW, Haltom JA, Albrecht YES, Lie T, Olali AZ, Widjaja GA, Ranshing SS, Angelin A, Murdock D, Wallace DC. SARS-CoV-2 mitochondrial metabolic and epigenomic reprogramming in COVID-19. *Pharmacol Res.* 2024; 204:107170.
- Houerbi N, Kim J, Overbey EG, Batra R, Schweickart A, Patras L, Lucotti S, Ryon KA, Najjar D, Meydan C, Damle N, Chin C, Narayanan SA, Guarnieri JW, Widjaja GA, et al. Secretome profiling reveals acute changes in oxidative stress, brain homeostasis, and coagulation following short-duration spaceflight. *Nature Communications.* 2024;15(1):4862.